

IUI Evaluation Report

Duolingo — Web Onboarding, Placement Test & Lesson Flow

Domain	Education / EdTech
Audience	Adult self-directed language learners using Duolingo's free web product
Persona	Aisha, 29, marketing coordinator learning French casually for an upcoming trip — motivated by streaks and quick wins, easily discouraged if the app doesn't already know what she's capable of
Input	16 screenshots — language picker, placement test, lesson flow, dashboard
Evaluated by	IUI Evaluator v1.1 — literature-grounded rubric
Date	14 August 2026
Score	19/35 (54%) Band: Developing IUI

IUI Maturity Band

Band	Raw	%
Pre-intelligent	0-10	0-28%
Emerging IUI	11-18	31-51%
Developing IUI (this score)	19-25	54-71%
Mature IUI	26-31	74-89%
Exemplary IUI	32-35	91-100%

Duolingo sits right at the entry threshold of Developing IUI: real adaptive and context-aware mechanisms are present and functioning, but the system explains almost none of its reasoning to the user, which caps it well below Mature.

Dimension Scores

#	Dimension	Score	Key Evidence
1	Adaptivity & Personalisation	3/5	Confirmed diagnostic — score meter fills as questions are answered, escalating difficulty (FR→EN to EN→FR) and determining Section 2 placement
2	User Modelling & Representation	3/5	Consequential user model with a visible downstream effect on placement; model itself not inspectable by the user
3	Context-Awareness	3/5	Temporal streak calendar (day-of-week) plus persistent Daily Quest task-state tracking
4	Transparency & Explainability	1/5	“Let's make this a bit harder...” is the only rationale ever given for an adaptive change
5	User Control & Controllability	3/5	“Start from scratch” opt-out at onboarding; Too Easy / Too Difficult / Report on every exercise

#	Dimension	Score	Key Evidence
6	Naturalness & Interaction Quality	3/5	Multimodal: speech recognition, audio playback, matching, word-bank construction
7	Proactivity & Task Support	3/5	Daily quests, streak-loss warnings, guidebook prompts — real but retention-oriented rather than learning-directed
Total		19/35	(54%)

What This Interface Does Well

The placement flow is genuinely adaptive rather than cosmetic. The “Find my level” path runs a confirmed diagnostic — a score meter that fills as questions are answered, escalating difficulty mid-test (the direction of translation flips from French→English to English→French, forcing active recall instead of passive recognition), landing the user in a specific section and unit. That's a legitimate user model with a visible downstream effect, distinct from the separate self-report question (“How much French do you know?”), whose own effect on placement remains unconfirmed.

The interface also respects user agency at the point of adaptation. “Start from scratch” sits as an equally-weighted alternative to “Find my level,” so a user who distrusts the adaptive placement can opt straight out. Inside lessons, the “Too Easy / Too Difficult / Report” row on every completed exercise gives a continuous, low-friction channel for contesting difficulty calibration.

Context-awareness is handled with more nuance than typical for a habit-loop app. The streak calendar is temporal (day-of-week, “starts tomorrow” framing) rather than a bare counter, and Daily Quests tie session-level progress to persistent, visible task state. Combined with speech input and audio playback, the interaction model is meaningfully multimodal for a language-learning context.

Priority Improvements (Top 3)

Improvement 1: Explain what the placement test actually measured

Dimension: Transparency & Explainability

Why it matters for Aisha: She sits through an adaptive diagnostic but is never told what determined Section 2 — or whether her self-reported answer played any role. If she expected to place higher, she has no way to understand or contest the decision.

Suggested approach: On the placement result screen, add a one-line rationale — e.g. “Placed in Section 2 based on your test score” — with a “why this level?” link for more detail.

IUI literature basis: Höök (1999) and Mezhoudi et al. (2015) identify transparency as paramount for adaptive systems; unexplained adaptation reads as arbitrary rather than intelligent.

BEFORE	AFTER
FR 10 Your Score connects course progress to real-life skills (no rationale given) CONTINUE	FR 10 e.g. "Placed in Section 2 based on your test score" → why this level? CONTINUE

Improvement 2: Surface the difficulty escalation instead of leaving it implicit

Dimension: Transparency & Explainability / Adaptivity

Why it matters for Aisha: The escalation is real (confirmed shift from recognition to recall), but “Let's make this a bit harder...” gives no indication of how. A learner who suddenly struggles can't tell whether it's her recall or the format that changed.

Suggested approach: Pair the escalation message with a concrete cue — e.g. a small “Now translating into French” tag on the question header.

IUI literature basis: Findlater & Gajos (2009) — adaptive mechanisms can improve one dimension (accuracy) while raising cognitive load along another (confusion) if the adaptation isn't legible in the moment.

BEFORE	AFTER
Great work! Let's make this a bit harder... (no cue what changes next) CONTINUE	Great work! Let's make this a bit harder... e.g. Now: English → French CONTINUE

Improvement 3: Give the self-reported proficiency question a visible purpose

Dimension: User Modelling & Representation

Why it matters for Aisha: “How much French do you know?” implies the system is modelling her, but its effect on the diagnostic that follows is never shown. An unused-seeming prompt erodes trust even when the rest of the model works.

Suggested approach: Either visibly blend the self-report into the diagnostic (e.g., fewer adaptive questions for someone who selects “I can have basic conversations”) or, if it's a segmentation input only, don't present it inside a flow that otherwise reads as diagnostic.

IUI literature basis: Brdnik, Heričko & Šumak (2022) identify representation — the system's model of the user being both accurate and legible — as a core IUI characteristic.

BEFORE	AFTER
How much French do you know? I'm new to French I know some common words I can have basic conversations answer's effect: unknown	How much French do you know? I'm new to French I know some common words I can have basic conversations e.g. 'This shortens your placement test'

Domain-Specific Commentary

For EdTech, the rubric weights User Modelling, Adaptivity, and Proactivity most heavily, since the core value proposition is scaffolding a learner's actual skill level rather than a fixed curriculum. Duolingo earns solid marks here: the adaptive placement mechanism (Adaptivity: 3, User Modelling: 3) is real and consequential, and Proactivity (3) — streak-loss warnings, quest bars, leaderboard-unlock prompts — genuinely keeps the learner returning, even though it's oriented more toward retention than toward diagnosing and closing specific skill gaps (e.g. no proactive weak-word review). For a learner like Aisha whose motivation is streak-driven, that's not a design flaw, but it does mean the intelligence on display is closer to a habit engine than a tutor.

Overall Verdict

Duolingo qualifies as a Developing IUI: it has a functioning adaptive core — a real diagnostic that changes question format based on measured performance, a user model with a visible downstream effect on placement, and multiple points where the user can steer or opt out of that adaptation — but it pairs that core with almost no transparency, leaving the user to infer or simply not know why the system did what it did. The gap between Developing and Mature here isn't more intelligence; it's explaining the intelligence that already exists.